

IoT-Based Smart Agriculture

System Design and Implementation Report

Automated Soil Moisture, Temperature and Humidity Monitoring
with Wi-Fi Based Cloud Connectivity and Smart Irrigation Control

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Platform: ESP32 Microcontroller

Project Category: IoT / Embedded Systems / Smart Agriculture

Abstract

This project presents the design and implementation of an IoT-based Smart Agriculture system that continuously monitors crop conditions using an ESP32 microcontroller interfaced with soil moisture, temperature, and humidity sensors. The sensor data is transmitted over Wi-Fi to a cloud platform, where it is displayed on a mobile and web dashboard for real-time visibility. Based on the measured soil moisture level, the system autonomously controls a water pump, enabling need-based irrigation rather than fixed-schedule watering. This reduces unnecessary water consumption, improves crop yield potential, and minimizes the manual effort required for farm monitoring. The system is designed to be low-cost, scalable, and suitable for both small farms and large agricultural fields.

1. Introduction

Agriculture remains one of the largest consumers of fresh water globally, and inefficient irrigation practices are a major contributor to water wastage. Traditional irrigation methods rely on fixed schedules or manual observation, which often leads to over-watering or under-watering of crops. This not only wastes a scarce resource but can also harm crop health and reduce yield.

The Internet of Things (IoT) offers a practical solution to this problem by enabling real-time, sensor-driven monitoring and automated decision-making. By continuously measuring soil moisture, temperature, and humidity, a smart agriculture system can determine the actual water requirement of the crop and irrigate only when necessary. This project implements such a system using an ESP32 microcontroller, environmental sensors, cloud connectivity, and an automated pump control mechanism.

2. Problem Statement

Manual irrigation and fixed-timer irrigation systems do not account for real-time soil and weather conditions. This results in:

- Excessive water usage due to over-irrigation.
- Crop stress and reduced yield due to under-irrigation.
- Lack of remote visibility into field conditions.
- High dependency on manual labor for monitoring and watering decisions.

3. Objectives

- Continuously monitor soil moisture, temperature, and humidity using low-cost sensors.
- Transmit sensor data wirelessly to a cloud platform using the ESP32's built-in Wi-Fi.
- Display real-time and historical sensor data on a mobile and web dashboard.
- Automatically control a water pump based on soil moisture thresholds.
- Reduce water wastage and improve overall irrigation efficiency.

4. System Architecture

The system follows a sense-transmit-process-act architecture. Sensors continuously sense the field environment, the ESP32 processes and transmits this data over Wi-Fi to a cloud platform, the cloud platform stores and visualizes the data on a dashboard, and finally the ESP32 acts on this data by controlling the irrigation pump through a relay module.

Stage	Function
Sensing	Soil moisture, temperature, and humidity sensors collect environmental data.
Processing	ESP32 reads sensor values, applies logic, and prepares data for transmission.
Communication	Data is sent over Wi-Fi (HTTP/MQTT) to a cloud platform.
Visualization	A mobile/web dashboard displays real-time and historical readings.
Actuation	Relay module switches the water pump ON/OFF based on soil moisture threshold.

4.1 System Block Diagram

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[ Soil Moisture Sensor ] [ DHT11/DHT22 Sensor ]
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 \ /
[ ESP32 Microcontroller ]
 | \
(Wi-Fi Module) (Relay Module)
 | \
[ Cloud Platform ] [ Water Pump ]
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[ Mobile / Web Dashboard ]
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5. Hardware Components

Component	Purpose
ESP32 Development Board	Main microcontroller with built-in Wi-Fi and Bluetooth for sensing, processing, and communication.
Soil Moisture Sensor	Measures volumetric water content in the soil (capacitive or resistive type).
DHT11 / DHT22 Sensor	Measures ambient temperature and relative humidity.
Relay Module	Electrically isolates and switches the water pump ON/OFF under ESP32 control.
Water Pump (5V/12V)	Delivers water to the crop when irrigation is triggered.
Power Supply	Provides regulated power to the ESP32 and connected peripherals.
Jumper Wires / Breadboard / PCB	Provides electrical connections between components.

6. Software and Tools Used

- Arduino IDE / PlatformIO — for writing and uploading ESP32 firmware.
- Embedded C / C++ — firmware programming language for the ESP32.
- Cloud Platform (e.g., Firebase, ThingSpeak, or Blynk) — for data storage and dashboard visualization.
- HTTP / MQTT Protocol — for lightweight communication between ESP32 and the cloud.
- Mobile/Web Dashboard — for real-time monitoring and manual override of pump control.

7. Working Methodology

- 1 The soil moisture sensor, temperature sensor, and humidity sensor are interfaced with the analog and digital pins of the ESP32.
- 2 The ESP32 periodically reads sensor values at a defined interval (e.g., every 5-10 seconds).
- 3 The ESP32 connects to a local Wi-Fi network and transmits the sensor readings to the cloud platform using HTTP requests or the MQTT protocol.
- 4 The cloud platform stores this data and updates the mobile/web dashboard in real time, allowing the user to remotely view soil moisture, temperature, and humidity trends.
- 5 The ESP32 continuously compares the soil moisture reading against a pre-defined threshold value.
- 6 If the soil moisture falls below the threshold (indicating dry soil), the ESP32 activates the relay module, which switches the water pump ON.
- 7 Once the soil moisture rises above the threshold (indicating sufficiently moist soil), the pump is automatically switched OFF, preventing over-irrigation.
- 8 The user can also monitor pump status and override control manually through the dashboard if required.

8. Results and Observations

The implemented system successfully monitors soil moisture, temperature, and humidity in real time and reflects this data accurately on the mobile/web dashboard with minimal latency. The automated pump control logic was observed to reliably switch the pump ON during dry soil conditions and OFF once adequate moisture was restored, demonstrating effective closed-loop irrigation control. Compared to manual or timer-based irrigation, the system showed a noticeable reduction in unnecessary water usage while maintaining consistent soil moisture levels suitable for crop health.

9. Advantages

- Reduces water wastage through need-based, sensor-driven irrigation.
- Minimizes manual monitoring effort with remote, real-time visibility.
- Improves crop health by maintaining optimal soil moisture levels.
- Low-cost and scalable hardware suitable for small and large farms alike.
- Historical data logging enables better long-term irrigation planning.

10. Applications

- Precision farming and smart irrigation for individual farms.
- Greenhouse and polyhouse automated climate and moisture control.
- Home gardening and urban rooftop farming automation.

- Large-scale agricultural estates requiring centralized remote monitoring.

11. Future Scope

- Integration of weather forecast data to further optimize irrigation scheduling.
- Addition of solar power and battery backup for fully off-grid operation.
- Use of machine learning models to predict irrigation needs based on historical trends.
- Expansion to multi-zone irrigation with independent control for different crop sections.
- Integration of soil NPK (Nitrogen-Phosphorus-Potassium) sensors for nutrient monitoring.

12. Conclusion

The IoT-Based Smart Agriculture system demonstrates a practical and cost-effective approach to modernizing traditional irrigation practices. By combining real-time environmental sensing, cloud-based monitoring, and automated pump control, the system reduces water wastage, lowers manual labor requirements, and supports healthier crop growth. The modular design allows for future enhancements such as predictive analytics and multi-zone irrigation, making it a strong foundation for scalable smart agriculture solutions.

13. References

- Espressif Systems, "ESP32 Series Datasheet," Espressif Systems Documentation.
- DHT11/DHT22 Sensor Datasheet, Aosong Electronics.
- Arduino Official Documentation, <https://www.arduino.cc/>
- ThingSpeak IoT Platform Documentation, <https://thingspeak.com/>
- Relevant IoT and Smart Agriculture literature on sensor-based irrigation automation.